

```

###Title of the Assignment: Data Visualisation I
###1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information
###about the passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see
###if we can find any patterns in the data.
###2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger
###is distributed by plotting a histogram.

```

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Suppress warnings for clean output
import warnings
warnings.filterwarnings('ignore')

# Set plot style
sns.set_theme(style='darkgrid')
plt.rcParams['figure.figsize'] = (8, 5)

```

```

# Load the built-in Titanic dataset
dataset = sns.load_dataset('titanic')

print(f"Shape: {dataset.shape}")
print(f"Rows: {dataset.shape[0]}, Columns: {dataset.shape[1]}")
dataset.head()

```

```

Shape: (891, 15)
Rows: 891, Columns: 15

```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```

# Dataset info
dataset.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   survived              891 non-null    int64
1   pclass                 891 non-null    int64
2   sex                   891 non-null    object
3   age                   714 non-null    float64
4   sibsp                 891 non-null    int64
5   parch                 891 non-null    int64
6   fare                  891 non-null    float64
7   embarked              889 non-null    object
8   class                 891 non-null    category
9   who                   891 non-null    object
10  adult_male            891 non-null    bool
11  deck                  203 non-null    category
12  embark_town           889 non-null    object
13  alive                 891 non-null    object
14  alone                 891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB

```

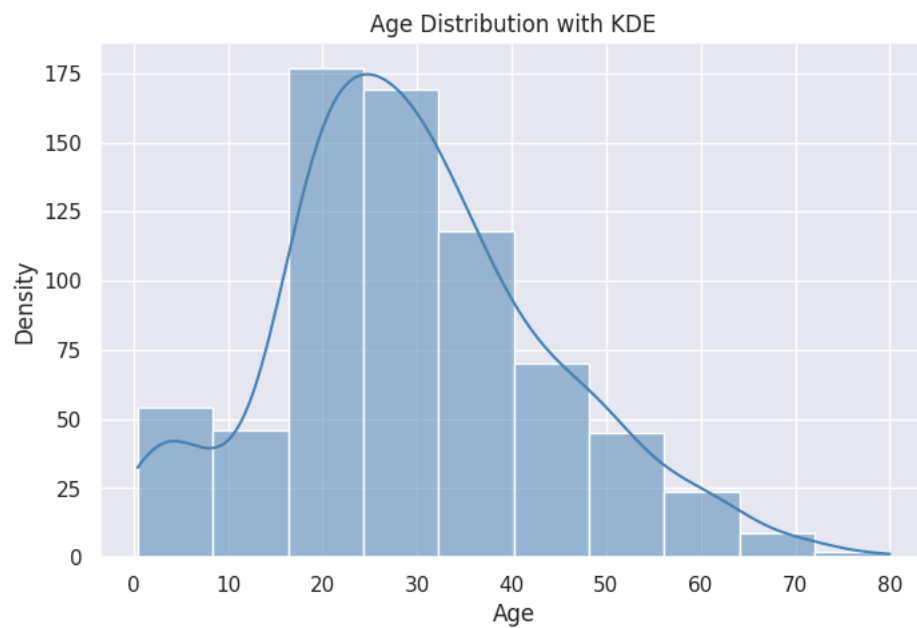
```

# Basic statistics
dataset.describe()

```

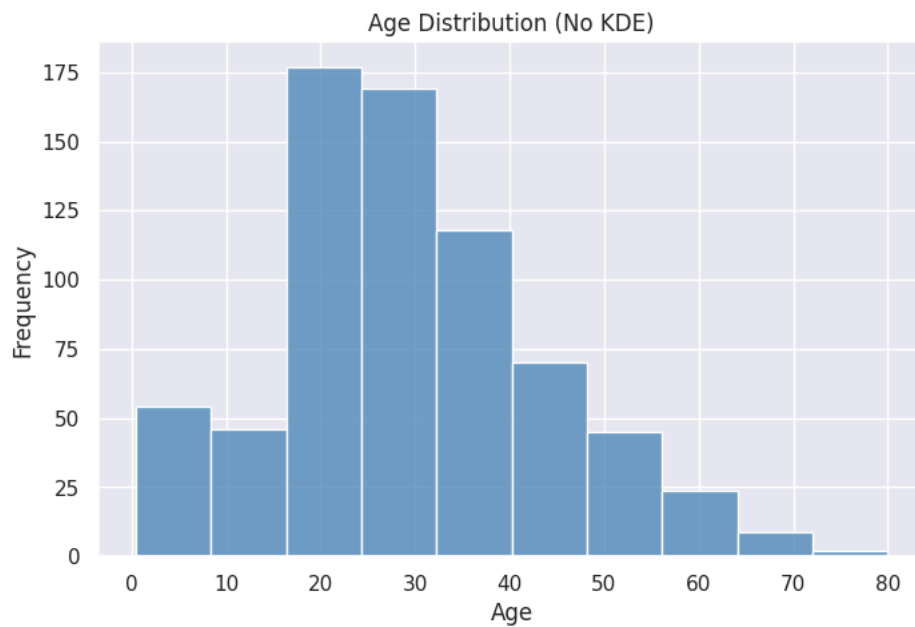
	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
# Distplot with KDE (using histplot as distplot is deprecated in modern seaborn)
plt.figure(figsize=(8, 5))
sns.histplot(dataset['age'].dropna(), bins=10, kde=True, color='steelblue')
plt.title('Age Distribution with KDE')
plt.xlabel('Age')
plt.ylabel('Density')
plt.show()
```



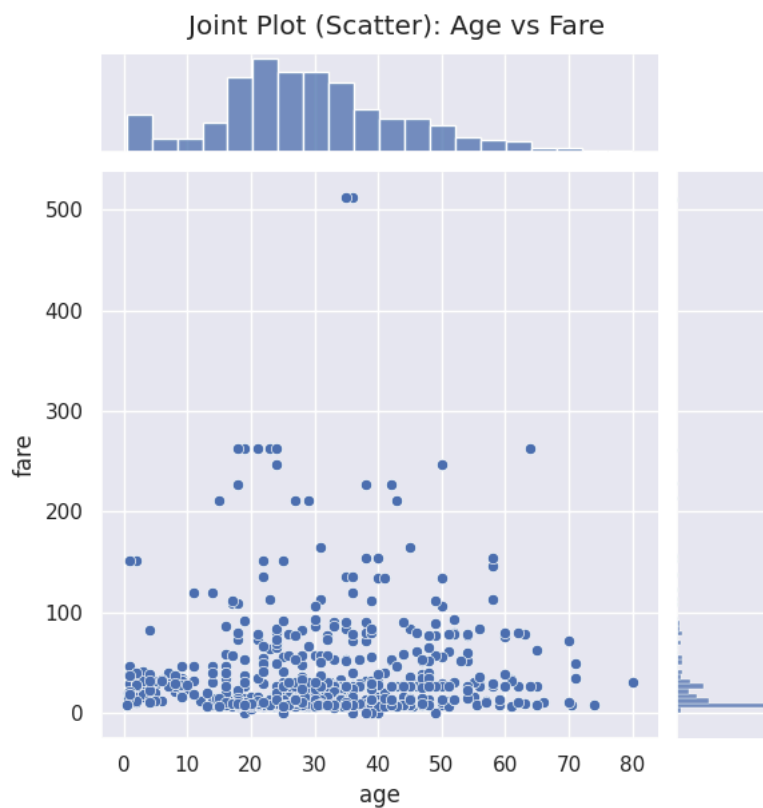
```
# Distplot without KDE
plt.figure(figsize=(8, 5))
sns.histplot(dataset['age'].dropna(), bins=10, kde=False, color='steelblue')
plt.title('Age Distribution (No KDE)')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()

print("Observation: ~50 passengers are aged 0-10, highest count is in the 20-30 age bracket.")
```



Observation: ~50 passengers are aged 0-10, highest count is in the 20-30 age bracket.

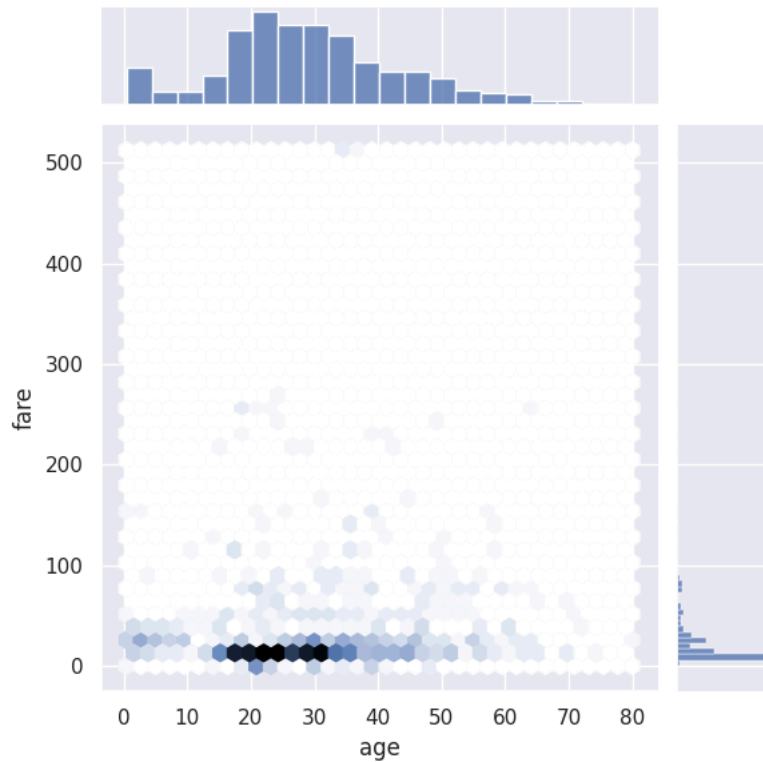
```
# Joint Plot – Scatter kind
g = sns.jointplot(x=dataset['age'], y=dataset['fare'], kind='scatter', height=6)
g.figure.suptitle('Joint Plot (Scatter): Age vs Fare', y=1.02)
plt.show()
```



```
# Joint Plot – Hex kind
g = sns.jointplot(x=dataset['age'], y=dataset['fare'], kind='hex', height=6)
g.figure.suptitle('Joint Plot (Hex): Age vs Fare', y=1.02)
plt.show()

print("Observation: Most passengers are aged 20-30 and paid fares between 10-50.")
print("No strong linear correlation between age and fare.")
```

Joint Plot (Hex): Age vs Fare

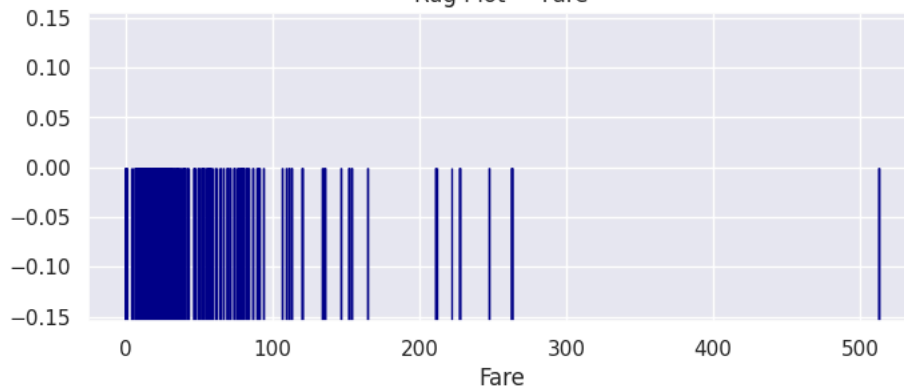


Observation: Most passengers are aged 20-30 and paid fares between 10-50.
No strong linear correlation between age and fare.

```
plt.figure(figsize=(8, 3))
sns.rugplot(data=dataset, x='fare', height=0.5, color='darkblue')
plt.title('Rug Plot — Fare')
plt.xlabel('Fare')
plt.show()

print("Observation: Most fare values are concentrated between 0 and 100.")
```

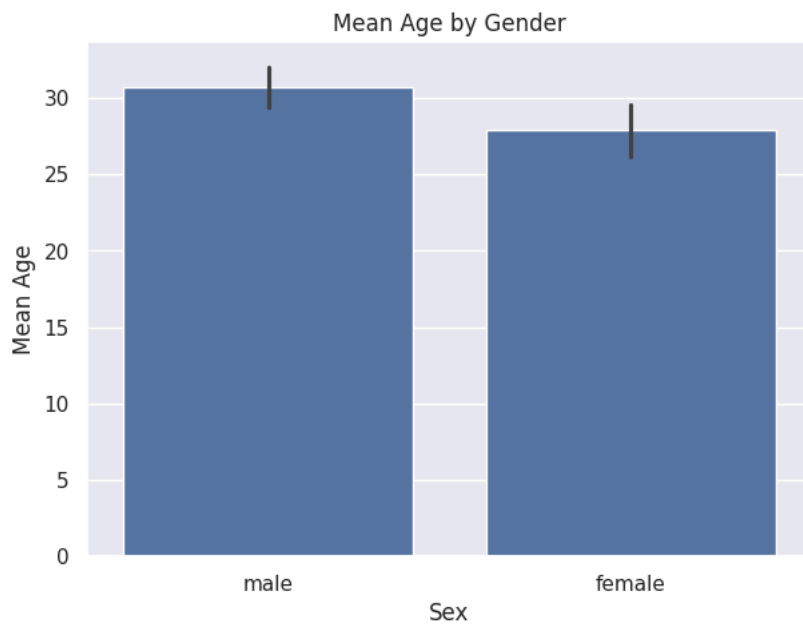
Rug Plot — Fare



Observation: Most fare values are concentrated between 0 and 100.

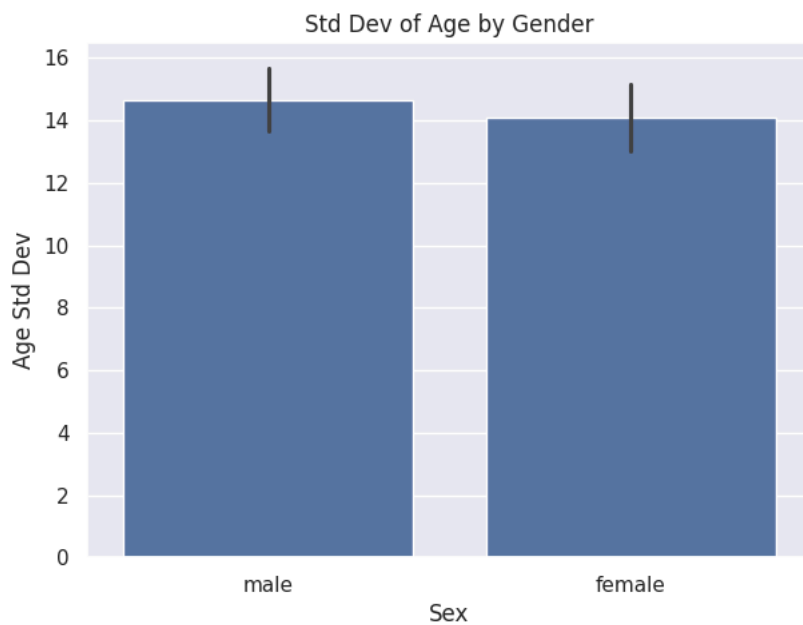
```
# Bar plot: Mean age by sex
plt.figure(figsize=(7, 5))
sns.barplot(x='sex', y='age', data=dataset)
plt.title('Mean Age by Gender')
plt.xlabel('Sex')
plt.ylabel('Mean Age')
plt.show()

print("Observation: Average age of male passengers ~30, female passengers ~28.")
```



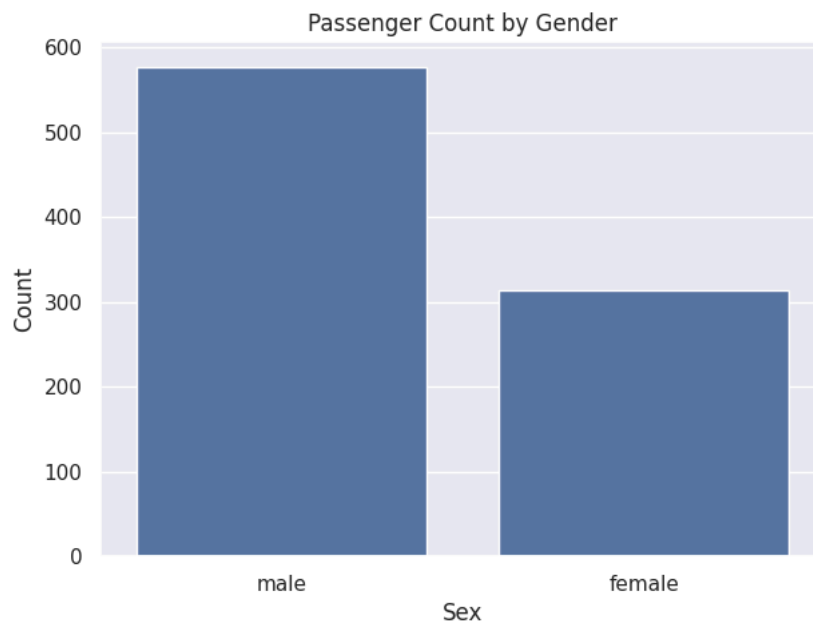
Observation: Average age of male passengers ~30, female passengers ~28.

```
# Bar plot: Standard deviation of age by sex (custom estimator)
plt.figure(figsize=(7, 5))
sns.barplot(x='sex', y='age', data=dataset, estimator=np.std)
plt.title('Std Dev of Age by Gender')
plt.xlabel('Sex')
plt.ylabel('Age Std Dev')
plt.show()
```



```
plt.figure(figsize=(7, 5))
sns.countplot(x='sex', data=dataset)
plt.title('Passenger Count by Gender')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.show()

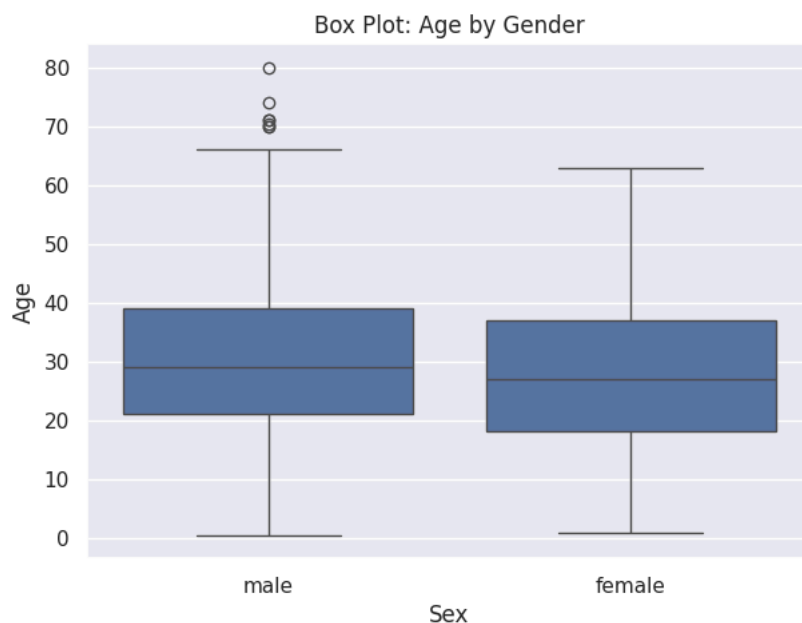
print(f"Male count : {dataset['sex'].value_counts()['male']}")
print(f"Female count: {dataset['sex'].value_counts()['female']}")
```



Male count : 577
Female count: 314

```
plt.figure(figsize=(7, 5))
sns.boxplot(x='sex', y='age', data=dataset)
plt.title('Box Plot: Age by Gender')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()

print("Box Plot Interpretation (Female):")
print(" Q1 (25%): ~1-20 years")
print(" Q2 (50%): ~20-28 years (median)")
print(" Q3 (75%): ~28-38 years")
print(" Q4 (100%): ~38-64 years")
print(" Dots above whiskers = outliers")
```

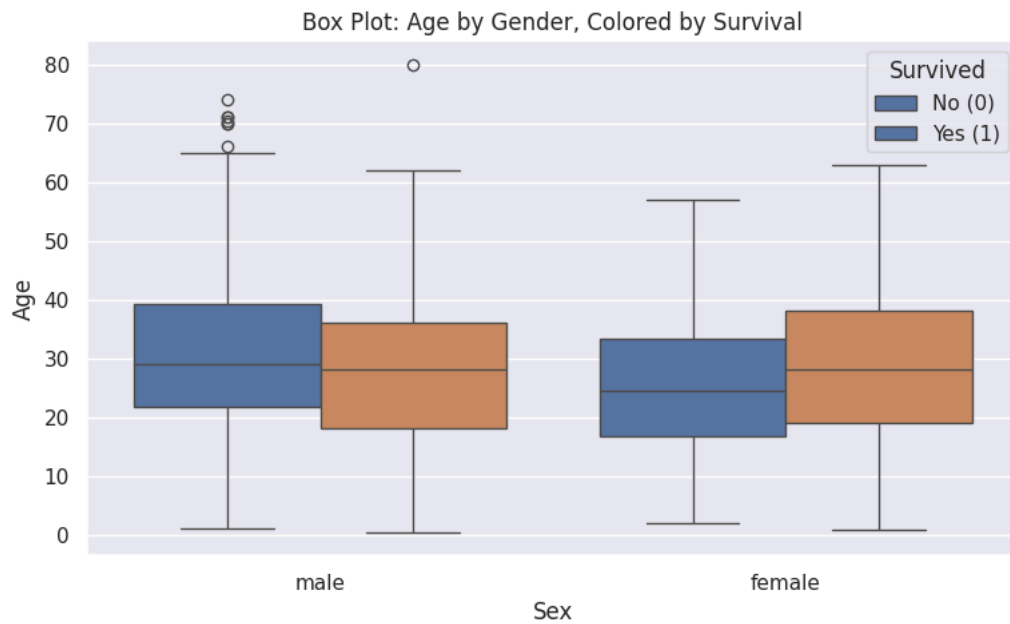


Box Plot Interpretation (Female):
Q1 (25%): ~1-20 years
Q2 (50%): ~20-28 years (median)
Q3 (75%): ~28-38 years
Q4 (100%): ~38-64 years
Dots above whiskers = outliers

```
# Box plot with hue (survival overlay)
plt.figure(figsize=(9, 5))
sns.boxplot(x='sex', y='age', data=dataset, hue='survived')
plt.title('Box Plot: Age by Gender, Colored by Survival')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.legend(title='Survived', labels=['No (0)', 'Yes (1)'])
```

```
plt.show()

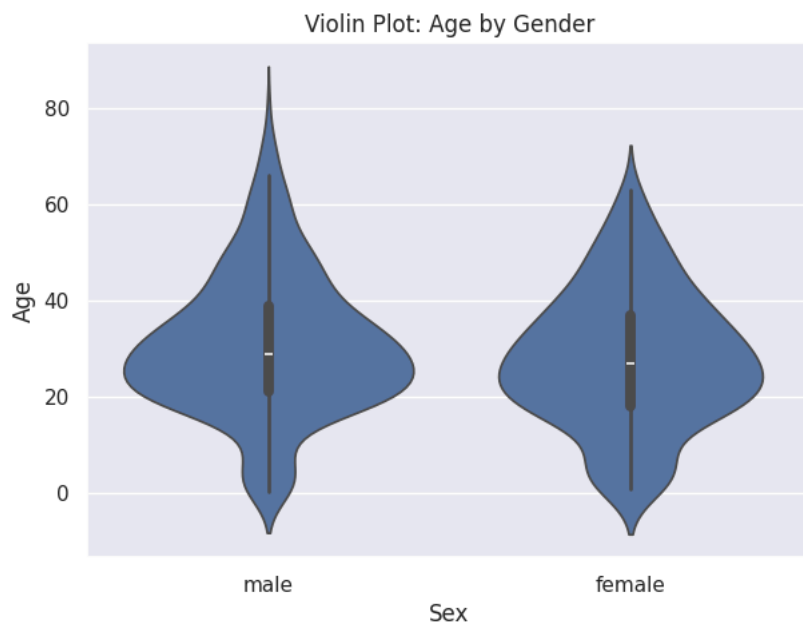
print("Observation: Among males, younger passengers had a higher survival rate.")
print("Female survival age variation is much wider for non-survivors.")
```



Observation: Among males, younger passengers had a higher survival rate.
Female survival age variation is much wider for non-survivors.

```
plt.figure(figsize=(7, 5))
sns.violinplot(x='sex', y='age', data=dataset)
plt.title('Violin Plot: Age by Gender')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()
```

print("Observation: The violin is thicker at 20-40 for males – higher passenger density in that range.")



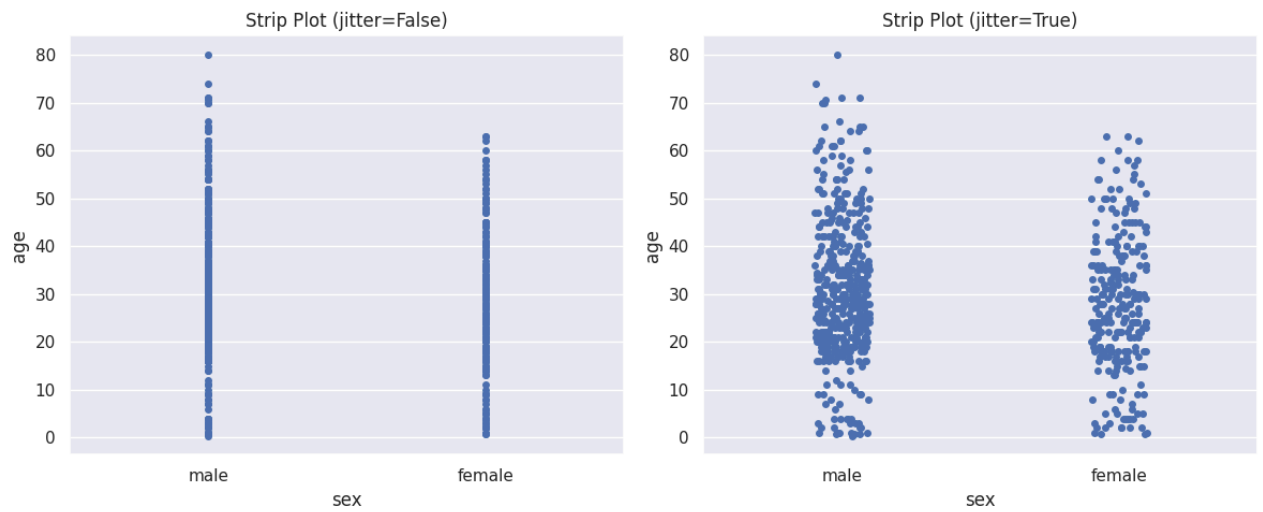
Observation: The violin is thicker at 20-40 for males – higher passenger density in that range.

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

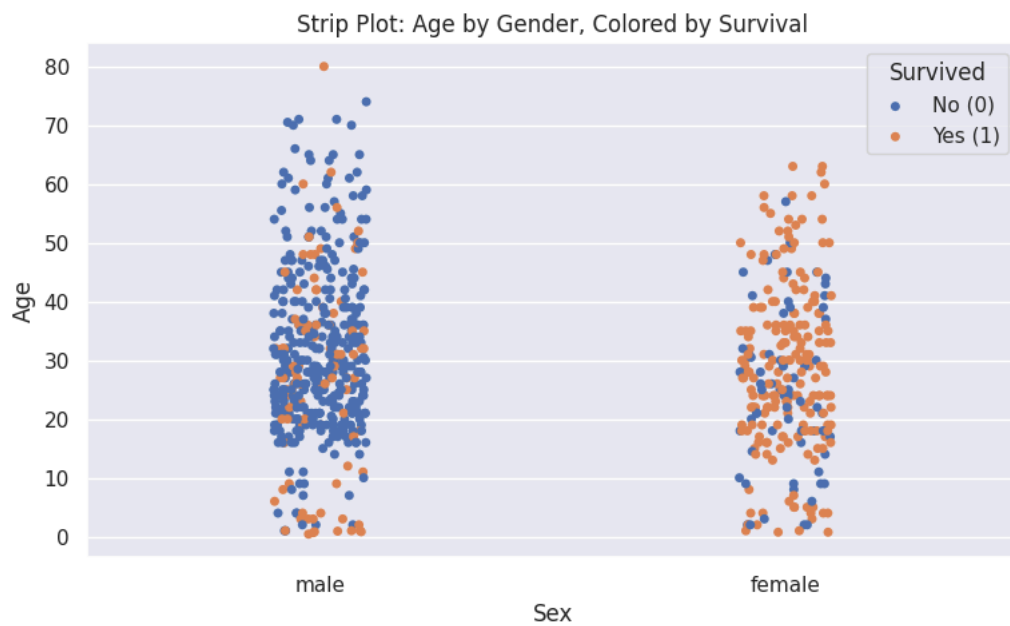
sns.stripplot(x='sex', y='age', data=dataset, jitter=False, ax=axes[0])
axes[0].set_title('Strip Plot (jitter=False)')

sns.stripplot(x='sex', y='age', data=dataset, jitter=True, ax=axes[1])
axes[1].set_title('Strip Plot (jitter=True)')

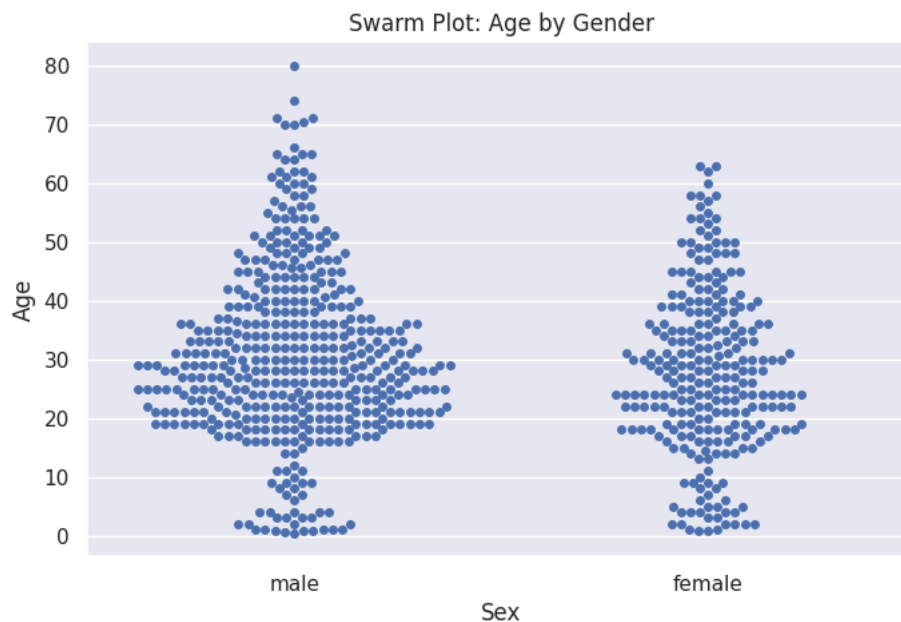
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(9, 5))
sns.stripplot(x='sex', y='age', data=dataset, jitter=True, hue='survived')
plt.title('Strip Plot: Age by Gender, Colored by Survival')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.legend(title='Survived', labels=['No (0)', 'Yes (1)'])
plt.show()
```

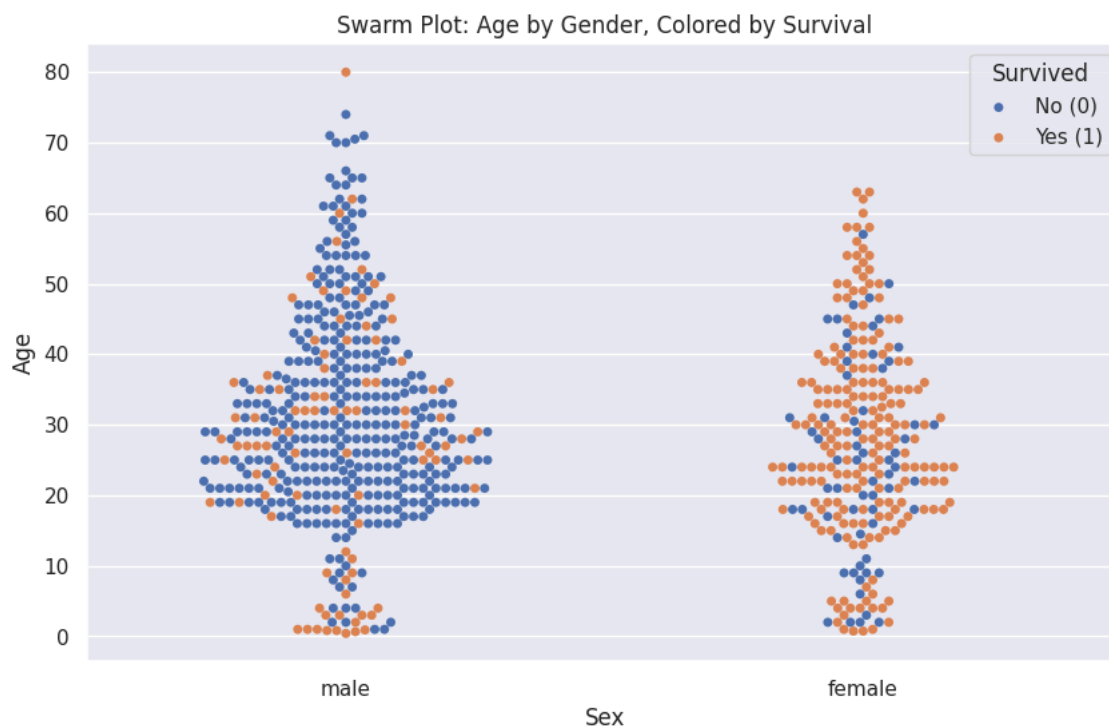


```
plt.figure(figsize=(8, 5))
sns.swarmplot(x='sex', y='age', data=dataset)
plt.title('Swarm Plot: Age by Gender')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()
```

```
plt.figure(figsize=(10, 6))
sns.swarmplot(x='sex', y='age', data=dataset, hue='survived')
plt.title('Swarm Plot: Age by Gender, Colored by Survival')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.legend(title='Survived', labels=['No (0)', 'Yes (1)'])
plt.show()

print("Observations:")
print(" - Male plot: more blue (non-surviving) than orange (surviving) → lower male survival ratio")
print(" - Female plot: more orange than blue → higher female survival ratio")
print(" - Male children (<10): more survivors than non-survivors")
```



Observations:

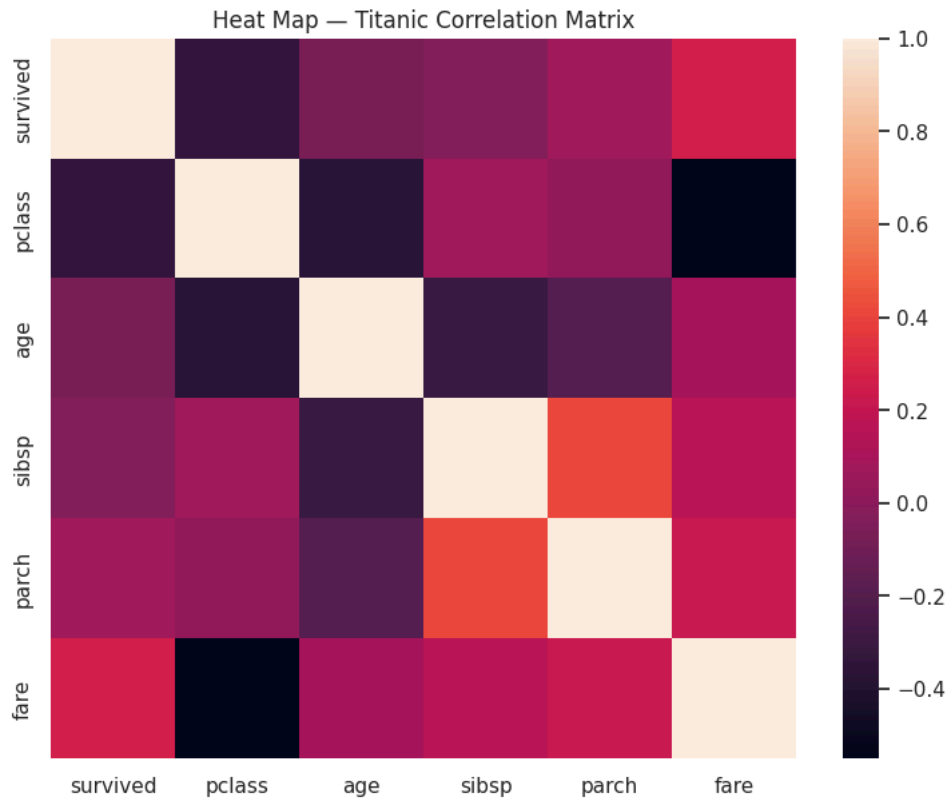
- Male plot: more blue (non-surviving) than orange (surviving) → lower male survival ratio
- Female plot: more orange than blue → higher female survival ratio
- Male children (<10): more survivors than non-survivors

```
# Compute correlation matrix (numeric columns only)
corr = dataset.select_dtypes(include='number').corr()
print("Correlation Matrix:")
corr
```

Correlation Matrix:

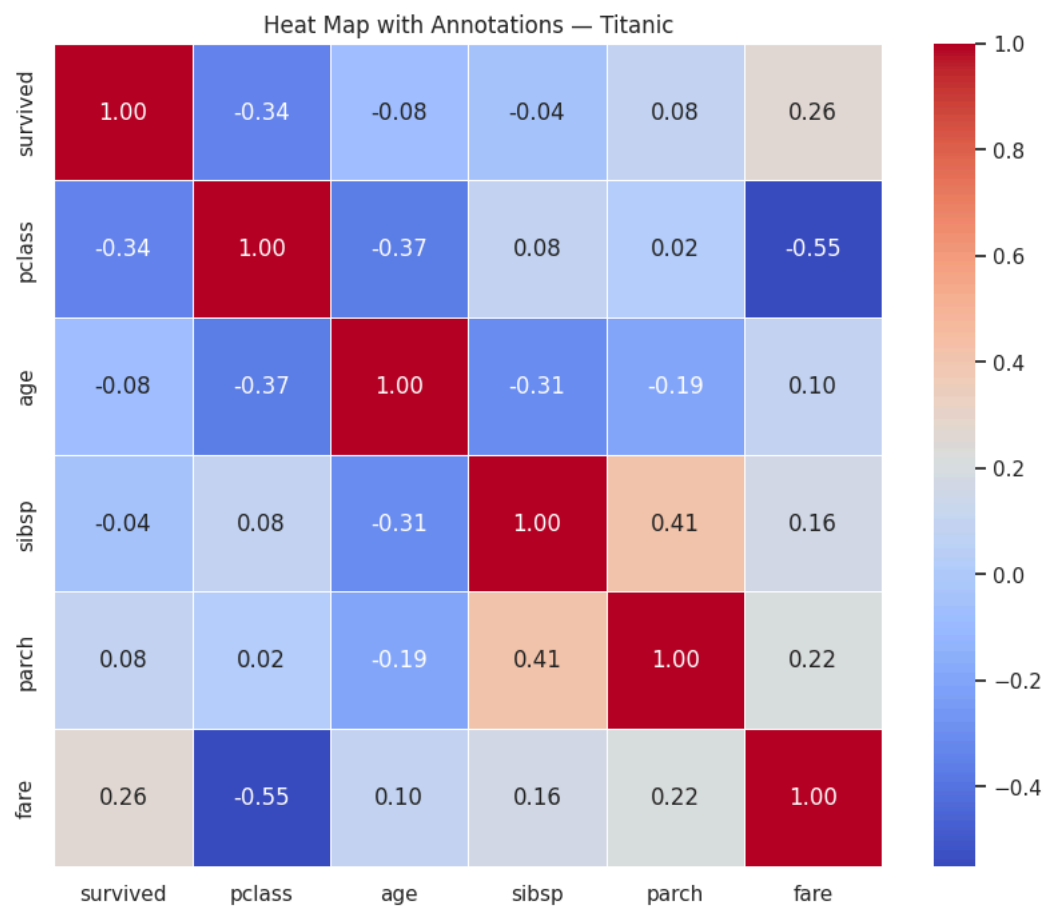
	survived	pclass	age	sibsp	parch	fare
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000

```
# Heat map without annotations
plt.figure(figsize=(9, 7))
sns.heatmap(corr)
plt.title('Heat Map - Titanic Correlation Matrix')
plt.show()
```

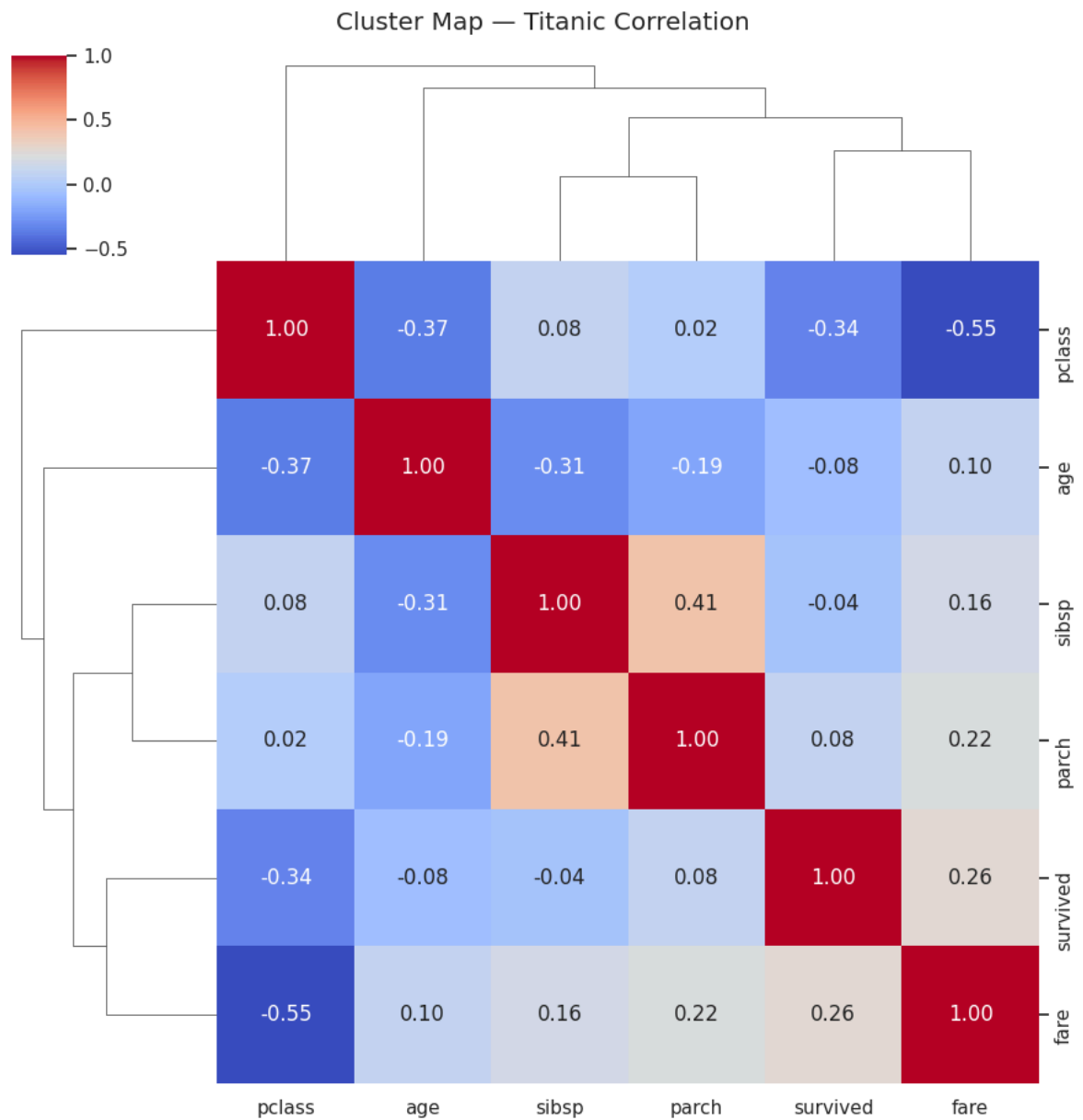


```
# Heat map with correlation values annotated
plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', linewidths=0.5)
plt.title('Heat Map with Annotations - Titanic')
plt.show()

print("Observation: 'fare' has moderate positive correlation (~0.26) with survival.")
print("'pclass' has a strong negative correlation (~-0.55) with fare - higher class = lower fare.")
```



```
sns.clustermap(corr, annot=True, fmt='.2f', cmap='coolwarm', figsize=(9, 9))  
plt.suptitle('Cluster Map — Titanic Correlation', y=1.02)  
plt.show()
```



```
# Basic histogram of fare
plt.figure(figsize=(9, 5))
sns.histplot(dataset['fare'], kde=False, bins=10, color='steelblue')
plt.title('Ticket Fare Distribution - Titanic')
plt.xlabel('Fare')
plt.ylabel('Count')
plt.show()

print("Observation:")
print(" ~730 passengers paid fares in the 0-50 range.")
print(" ~100 passengers paid fares around 50-100.")
print(" Very few passengers paid above 300 (extreme outliers - luxury cabins).")
```

Ticket Fare Distribution — Titanic

